

REPORT NORTHSTONE 160323

From my visit to the NORTHSTONE (Antrim) factory from 14 to 16 March 2023, I give my assessment and recommendations on the different equipment and processes that I have been able to review.

SAND HOPPERS / BELT CONVEYORS, CEMENT SILOS, PIGMENT DISPENSER

Very good equipments and installation.

MIXER:

R24 ERICH, it's a very good mixer.

LIQUID TO AVOID CONCRETE INCRUSTATIONS IN THE MIXER

Once the inside of the mixer has been cleaned, it is advisable to impregnate this liquid (**ZETOLAN-MEK 3**) with the exterior of all the elements that may come into contact with the concrete of the mixer, thereby avoiding the incrustations and the ones that are generated can be easily cleaned



ZETOLAN MEK 3



SKU/CODE: MEK3
PROTECTION FOR CONCRETE MIXERS AND CONSTRUCTION EQUIPMENT
ZETOLAN-MEK3 forms an effective adhesive protective film on the surfaces on which it has been sprayed, preventing the adhesion of residual deposits of concrete and mortar. Deposits can be removed by means of a water jet. As soon as the protective film wears off, a new protective layer of ZETOLAN-MEK3 can be applied to the wet surface after cleaning.
Advantages:
Costly contamination of essential parts and grease grooves is avoided.
Longer service life of the machinery due to better care.
Reduces labour cost for cleaning.
Does not damage rubber, glass, metal, paint, wood or clothes.
Economical application, since only a thin film is needed for protection.

Imágenes de zetolan mek



METAL DETECTOR

I recommend the use of a metal detector, the closed loop metal detector is the best for this application. To detect a 10 mm nut, to avoid breakage of the slipper, roller or mould, it must be tested every week. or mould, it is necessary to test its operation every week



PALLETS ALUMINIUM :

Capacity pallets, 7 chambers 39.200 DONARD, 3 chambers 16.800 CAUSEWAY, 4 chambers 22.400 DERRIE pallets.

Side moulds all very clean and on the surface with some easy to clean dirt.

On the top of the DONARD mould some small concrete adherence, no problema.



It also has wear on the corner due to contact with the mould release pallete blades.



No pallet side scrapers act, if there are no concrete adherence they are not necessary.



PALLETS FLOW INVERTER.

The polycords of the lower outlet conveyor are quite damaged, so it would be necessary to consider the replacement

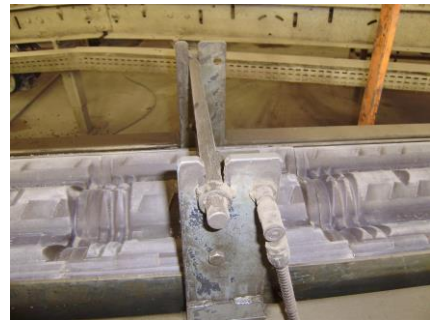


CONVEYORS LINE PALLETS FLOW INVERTER TO EXTRUDER

I recommend to identify wear of the guides, polycords and pulleys, in order to evaluate the replacement before a production line shutdown.

FLAPS PALLETS

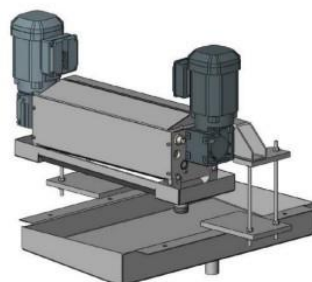
Based on comments in profile changes on the accommodation or adjustment of the flap, I recommend, possible change of the screen system for each model to a fixed non-adjustable one.



OIL APPLICATOR.

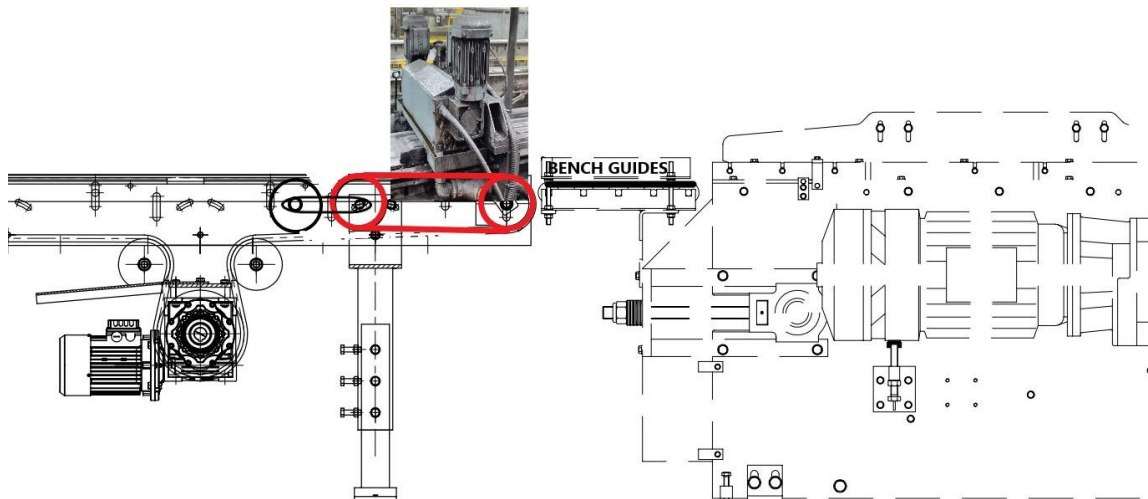


***Fit anti-fall oil reservoirs ,release agent in application system.**



POSITION OF THE OIL APPLICATOR.

I recommend the study and evaluation to move the oil applicator forward, at the start of the infeed conveyor, making the conveyor area independent of the applicator and replacing the infeed rollers with a base with bench guides to allow the pallets to move at the infeed of the pusher chain.



PALLETS PUSHING SYSTEM

It is necessary to check daily the areas where the carriages are pushed to the pallet and their bearings.

It is also advisable to clean the chain with "low pressure air", but "not to grease it".

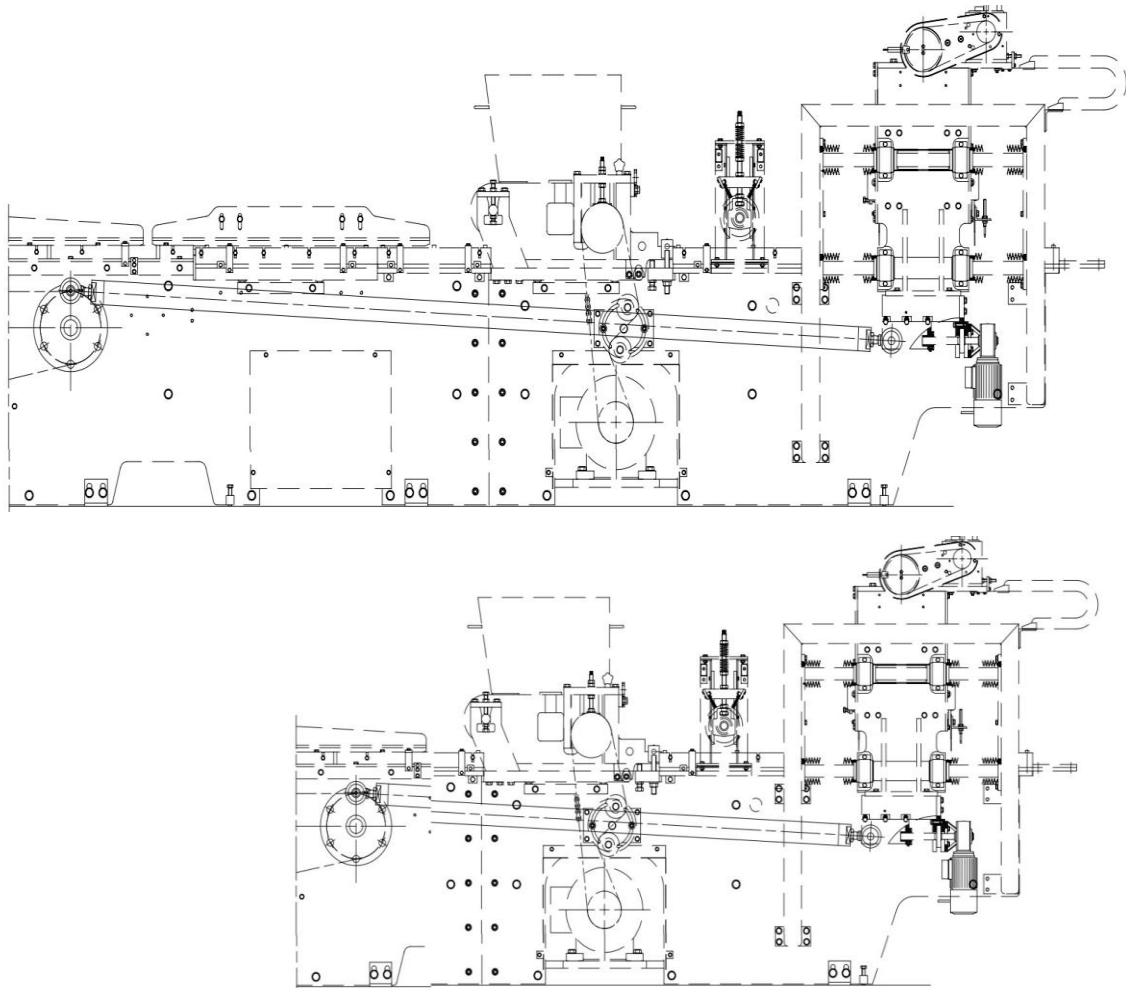
Check monthly ¿? the distance between pusher fingers, today it measures 421 mm.

POSITION OF THE PALLET PUSHING SYSTEM

Given the changes of guides and adjustments in the use of the 3 tile profiles every day and the future non-use of the space between the pushing system and the extruder head, I recommend the study and evaluation with VORTEX to move the pushing system forward by 1,260 mm.

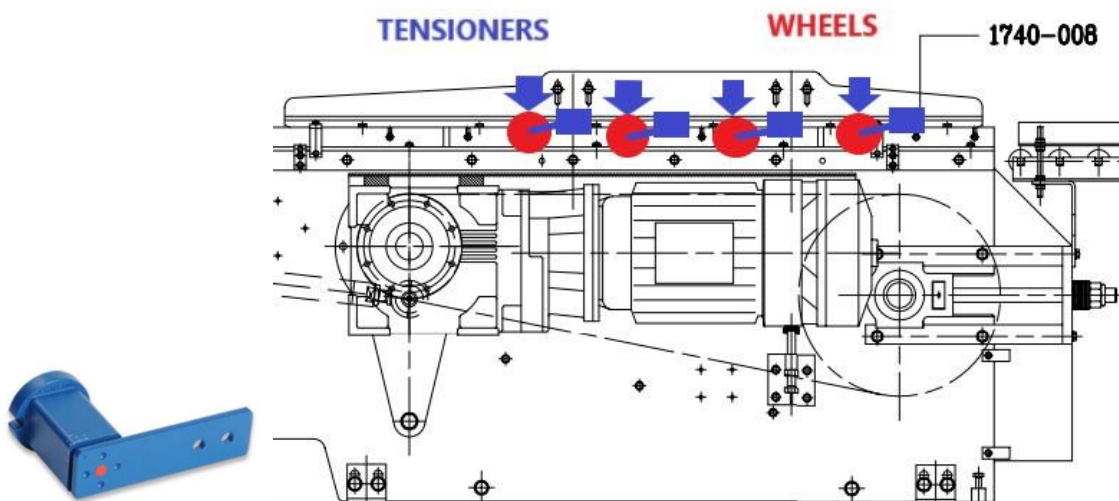
This would eliminate the anti-lifting system of the pallets, reduce the bed lateral guides and reduce the length of the bar that moves the cutting system.

Temporarily, a base with guides could be placed in this area so that the pallets only move along the base and laterally guided to the entrance of the extruder head.



ANTI-LIFTING SYSTEM OF THE PALLETS

At the moment it does not work with the DONARD / CASEWAY pallets, therefore, with the use of the 3 profiles, I recommend the study and evaluation of the possible placement of polyurethane wheels with tensor that circulate through the valley areas of the DERRIE pallet, which has little difference in height with the bases of the DONARD / CASEWAY



EXTRUDER HEAD

The adjustment values of the DONARD / CAUSEWAY models with tungsten (TC) rollers and slippers have been identified with the factory operators, as well as with the DERRIE model which uses steel rollers and slippers.

I recommend the study and evaluation of modifying one area of the CAUSEWAY tile interlocking from straight to curved.



HOPPER EXTRUDER.

I recommend increasing the concrete column in the extruder, to guarantee the feeding of concrete to the roller and slipper.



CUTTING SYSTEM

Study and evaluate the possible application of a frontal knife water and air nozzle (currently only lateral) in the production of CAUSEWAY tiles on the rounded cut area.

Check the protections of the belt that drives the cutting blades, as it vibrates a lot and causes excessive noise.



POLYCORD CONVEYORS AT THE EXTRUDER TO TWO RACKERS .

LOADING CONVEYOR FOR TWO RACKERS

To identify wear of the guides, polycords and pulleys, in order to evaluate the replacement before a stop of the production line.

RACKERS (4 TILES - 35 FLOORS)

Some racks need to be checked for cleanliness and deterioration, the paint on the top guides is lifting off some of them.

DOUBLE LOADING PUSHER (76.2 PITCH) FOR RACKS PLANT

It is necessary to fit rubber buffers as a stopper at the front and rear stops of the 2 pushers or reduce current shocks.

RACKER - 35 FLOORS

All guides must be checked for cleanliness and deterioration.

Check the tensioning to be equal and the chain measurement to check its elongation initially the measurement corresponds to (102 " = 2590,8 mm).

Checked the 150323 the measurement is 2595 mm (+4.2 mm), in the 4 chains of the rackers.

The initial position of the front racker is higher than the rear racker (+ 8 mm), it is interesting to follow up.

ROLLER-WAY FOR RACKS CIRCULATION IN THE CHAMBERS

EXTRACTOR FOR RACKS

RACKS LOADING/UNLOADING ROLLER-WAYWET TILES RACKS TRANSFER UNIT AND

TRANSFER CAR LOADING

RACKS ADVANCEMENT GROUP

DRY TILES TRANSFER CAR UNLOADING AND RACK TRANSFER TO THE UNLOADING LINE

WET TILES RACKS PUSHER IN THE CHAMBERS

The positioning of the racks in their stations must be ensured, especially in the loading and unloading positions of the fresh and dry tile racks.

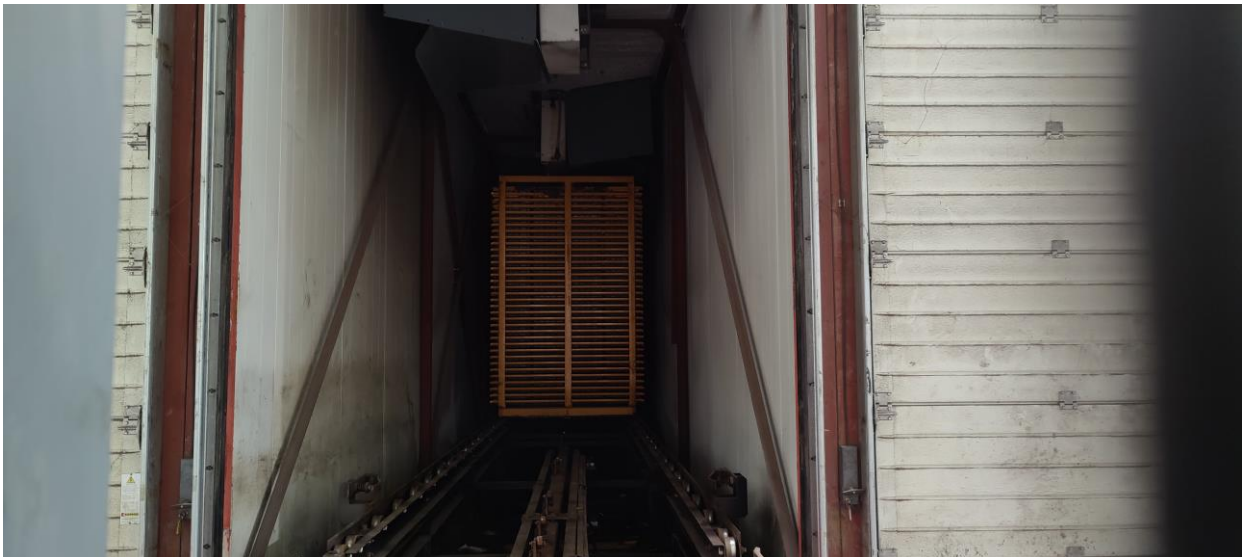
DRY AND WET TILES TRANSFER CARS

In my opinion, the fresh tile transfer has some shaft / bearing problems in some of the wheels, it makes a noise out of the ordinary compared to the cured tile transfer.



14 CURING CHAMBERS

I do not have the time to evaluate the water in the base of the chambers, nor to assess the current curing cycles.



UNLOADING PUSHER

It is necessary to fit rubber buffers as a stopper at the front and rear stops of the 2 pushers or reduce current shocks.

TILES DERACKER (4 TILES - 35 FLOORS)

All guides should be checked for cleanliness and deterioration.

Check tensioning to ensure that it is equal and the chain measurement to check its elongation chain, the initially the measurement corresponds to (102 " = 2590.8 mm).

Checked the 150323 the measurement is 2,602 mm (+11.8 mm), on the 2 deracker chains.

It was not possible to check the initial position because there was no racker to check the difference in height and make a follow-up.

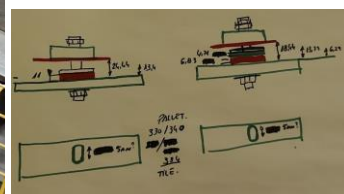
L= 5000 UNLOADING CONVEYOR

Identify wear of the wheels, polycords and pulleys, in order to evaluate the replacement before a stop of the production line.

"T150" DEPALLETER



Regarding the adjustments of the depalletter machine for the 3 tile profiles, depending on the current pallet conveyor, the only thing I would consider is a study and evaluation of the accommodation of a pre-depalletter unit at the entrance of the current conveyor, with guides and adjustable for each model.



POLYCORD CONVEYORS FROM DEPALLETER TO PALLETS FLOWINVERTER.

Identify wear of the guides, polycords and pulleys, in order to evaluate the replacement before a stop of the production line.

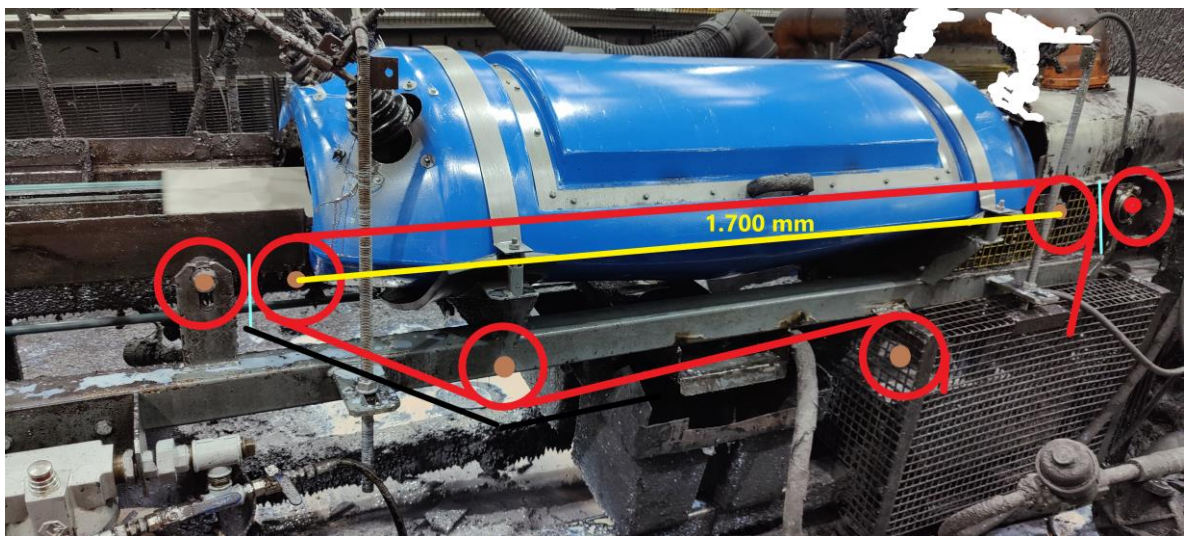
POLYCORD CONVEYORS FROM DEPALLETER TO PAINT APPLICATOR.

Identify wear of the guides, polycords and pulleys, in order to evaluate the replacement before a stop of the production line.

Depending on the contamination of water with paint on the conveyor sections throughout the area, I would consider the study and assessment of reducing the length of the conveyor where the paint is applied to the minimum necessary.

Applying the paint “only” at the entrance of the conveyor to help reduce the length of the conveyor and distance the operator's application from the selection of tiles.

Redirect all conveyor cleaning water /Paint discharges to a central tank.



Delimit the area of possible contamination on the ground with a 5 / 10 cm high fence.



POLYCORD CONVEYORS FROM PAINT APPLICATOR TO DRY ROTARY.

Identify wear of the guides, polycords and pulleys, in order to evaluate the replacement before a stop of the production line.

DRYING ROTARY 40 FLOORS

Heavy wear on all the guides, they need to be replaced.



RACKER 50 FLOORS

All guides must be checked for cleanliness and deterioration.

Check the tensioning to be equal and the chain measurement to check its elongation.

UNLOADING PUSHER DRYING ROTARY

DOUBLE LOADING PUSHER DRYING ROTARY (DONARD / CASEWAY)

It is necessary to fit rubber buffers as a stopper or reduce current shocks

MAINTENANCE.

The following are some of the maintenance and efficiency assessments that I'm sure some of them apply to this factory, but I always like to point them out.

I don't know the planning and maintenance history, in relation to the equipment must be high, preventive maintenance is necessary and the lubrication routes.

Each operator should carry out a minimum level of preventive maintenance and, depending on the level of knowledge, some basic corrective maintenance work.

As a general rule in the factories that I have worked in, to achieve and keep high efficiency, by 1 hour of production needs 1 hour of maintenance at the end of the daily production.

When the time of a production stoppage is more than 15 minutes, it should be necessary to study it and monitor it in order to propose a solution or to avoid the occurrence of this high stoppage time.

In order to increase efficiency, it is necessary to provide a solution to the continuous stoppages of the installation, with the identification of the stoppages, their analysis and possible improvements that solve these stoppages with the participation of the entire production team of the factory.

"5 whys" It is the problem-solving method used by exploring the underlying cause and effect of certain problems.

The main objective is to determine the root cause of a defect or problem by successively asking the question "Why?"

SUMMARY

This factory produces very good quality tiles, the efficiency can be improved with more maintenance and solving the problems of the current shutdowns.

From my experience with 3 daily profiles, the average efficiency loss is 9 to 12 %.

I recommend 2 production shifts, with a total of 16 hours of production and maintenance 2 officers at night 16 hours.

To keep a high efficiency with a production of 20 hours for 5 days, you need a minimum of 2 cleaners 8 hours and 2 mechanics 8 hours at night and maintenance on Saturday 8 hours 5 mechanics, with the possible support of an external workshop.

I hope that my suggestions and advice can be useful, to increase the efficiency of the factory.

Antrim 160323